

CE 310 — Week 10 Assignment: Multiple Linear Regression

Week: 10 | **Topic:** Multiple Linear Regression — Diagnostics and Application

Points: 60 pts total (3 problems)

Due: Wed Nov 4, 9:00 AM — submit to D2L

Overview

The in-class exercise built multiple regression models adding BusinessHours and Season as predictors alongside Outdoor_Temp_F. This assignment asks you to extend that work: diagnose which predictors actually matter, apply the interaction model to track-specific design problems, and reflect critically on when model complexity is (and is not) worth the trade-off.

You will need: your completed exercise notebook (Wk10.03_In_Class_Exercise.ipynb), the DOE dataset (Week5_DOE_MedOffice_ZoneTemp_2023.csv), and a Python environment (Google Colab or local).

How to Submit

Read this before you submit. Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	Download as .ipynb — File → Download → Download .ipynb (not PDF, not .py)
Major declaration	Set MAJOR = "CE" or MAJOR = "Arce" in the Setup cell — this determines which Problem 2 track is graded
Print lines	Do not edit or delete any <code>print(f"ANSWER_... = ...")</code> line
Written responses	Put each in its own markdown cell immediately after the code cell it responds to
Name and NetID	Fill in NAME and NETID in the identity cell and run it — that cell is what matches your work to your student record. A blank NETID takes a 5-point deduction.
File name	CE310_W10_Assignment_<NetID>.ipynb e.g. CE310_W10_Assignment_abc123.ipynb
Submission	Upload the .ipynb file to D2L → Assignments by 9:00 AM, Wed Nov 4

Setup cell (run first):

```
NAME = "" # e.g. "Jordan Reyes"
NETID = "" # e.g. "abc123"
print(f"NAME = {NAME}")
print(f"NETID = {NETID}")

import pandas as pd
import statsmodels.formula.api as smf
```

```
MAJOR = "CE"    # or "ArcE" – declares your Problem 2 track
```

```
df = pd.read_csv("Week5_DOE_MedOffice_ZoneTemp_2023.csv")
df['BH'] = (df['BusinessHours'] == 'Yes').astype(int)
print(f"MAJOR = {MAJOR}")
```

Problem 1 — Model Diagnostics and Extension (30 pts)

Do not repeat the exercise computations. Use the model coefficients, R^2 values, and residual plots from your completed exercise notebook as starting points.

P1a (8 pts) — Why Season Adds Almost Nothing

In the exercise (Model 3), adding $C(\text{Season})$ alongside Outdoor_Temp_F increased R^2 from 0.601 (simple regression) to only 0.611 — a gain of just 1 percentage point. Yet adding BH alone (Model 2) increased R^2 from 0.601 to 0.697 — a gain of nearly 10 percentage points.

```
season_means = df.groupby('Season')['Outdoor_Temp_F'].mean()
ANSWER_P1a_summer_mean = ____
ANSWER_P1a_winter_mean = ____
print(f"ANSWER_P1a_summer_mean = {ANSWER_P1a_summer_mean}")
print(f"ANSWER_P1a_winter_mean = {ANSWER_P1a_winter_mean}")
```

In approximately 75 words, in a markdown cell, explain **why Season adds so little predictive power** once Outdoor_Temp_F is already in the model. Your explanation must: 1. Use the word **collinearity** (or collinear). 2. Reference the two numbers you just printed.

Graded on accuracy of the collinearity explanation and correct use of the quantitative example.

P1b (8 pts) — DayType as an Alternative Occupancy Proxy

Instead of BusinessHours, fit a model using DayType (Weekday vs. Weekend) as the second predictor:

```
model_dt = smf.ols('Cooling_kWh ~ Outdoor_Temp_F + C(DayType)', data=df).fit()
ANSWER_P1b_r2 = ____
ANSWER_P1b_weekend_coef = ____
ANSWER_P1b_weekend_pvalue = ____
print(f"ANSWER_P1b_r2 = {ANSWER_P1b_r2}")
print(f"ANSWER_P1b_weekend_coef = {ANSWER_P1b_weekend_coef}")
print(f"ANSWER_P1b_weekend_pvalue = {ANSWER_P1b_weekend_pvalue}")
print(model_dt.summary())
```

In a markdown cell: Is DayType a statistically significant predictor after controlling for outdoor temperature (justify using the p-value)? Compare this model's R^2 to Model 2 ($R^2 = 0.697$, which used BH). Which predictor — BusinessHours or DayType — explains more variance in cooling energy? Why might one outperform the other? (Hint: think about what DayType captures vs. what BusinessHours captures at the hourly level.)

P1c (7 pts) — Adding Both BH and DayType

Fit a model with both BH and DayType:

```
model_bh_dt = smf.ols('Cooling_kWh ~ Outdoor_Temp_F + BH + C(DayType)', data=df).fit()
ANSWER_P1c_r2 = ____
ANSWER_P1c_adj_r2 = ____
print(f"ANSWER_P1c_r2 = {ANSWER_P1c_r2}")
print(f"ANSWER_P1c_adj_r2 = {ANSWER_P1c_adj_r2}")
print(model_bh_dt.summary())
```

In a markdown cell: Compare to Model 2 ($R^2 = 0.697$, Adj. $R^2 = 0.697$). Does adding DayType on top of BH improve R^2 ? Does Adj. R^2 increase or decrease? What do you conclude about the **incremental information** in DayType once BH is already included? (Hint: consider what BH and DayType each capture — are they measuring similar or different aspects of occupancy?)

P1d (7 pts) — Interaction Model Predictions at the 0.4% Design Temperature

Using Model 5 (the interaction model $\text{Cooling_kWh} \sim \text{Outdoor_Temp_F} * \text{BH}$) from the exercise, predict the cooling demand at the **0.4% design temperature = 103.8°F** for two scenarios:

```
model5 = smf.ols('Cooling_kWh ~ Outdoor_Temp_F * BH', data=df).fit()
b0, b1, b2, b3 = (model5.params['Intercept'], model5.params['Outdoor_Temp_F'],
                  model5.params['BH'], model5.params['Outdoor_Temp_F:BH'])
```

```
design_temp = 103.8
```

```
# (a) Occupied scenario (BH = 1)
```

```
ANSWER_P1d_occ_pred = ____
print(f"ANSWER_P1d_occ_pred = {ANSWER_P1d_occ_pred}")
```

```
# (b) Unoccupied scenario (BH = 0)
```

```
ANSWER_P1d_unocc_pred = ____
print(f"ANSWER_P1d_unocc_pred = {ANSWER_P1d_unocc_pred}")
```

```
# Compare to Model 2 prediction at the same temperature
```

```
model2 = smf.ols('Cooling_kWh ~ Outdoor_Temp_F + BH', data=df).fit()
ANSWER_P1d_model2_pred = ____
```

```
    model2.params['Intercept'] + model2.params['Outdoor_Temp_F']*design_temp + model2.params['B
```

```
)
ANSWER_P1d_diff = ____
```

```
print(f"ANSWER_P1d_model2_pred = {ANSWER_P1d_model2_pred}")
print(f"ANSWER_P1d_diff = {ANSWER_P1d_diff}")
```

State the four Model 5 coefficients by name (b_0 – b_3 above) before you substitute — show the full substitution step-by-step in a markdown cell, not just the code.

In the same markdown cell: at the 0.4% design condition, which model would lead to a larger-capacity chiller specification? Explain in one sentence why the interaction model's occupied estimate is higher than the additive model's estimate at high temperatures.

Problem 2 — Track Application (18 pts)

Complete only the track matching your declared MAJOR. CE → Track A. ArcE → Track B.

Track A — CE: Can the Campus Loop Absorb a New 24/7 Load? (18 pts)

You are on the campus utilities team. A new **24/7 facility** is to be connected to the existing district chilled-water loop, and the question in front of the capital planning committee is whether the central plant can carry it or whether the campus has to fund added capacity. The facility runs around the clock, so it is always “occupied” in our model (BH = 1 at all hours).

This is a capacity question about shared infrastructure, not an equipment selection — the plant already exists, and what you are deciding is whether it has room.

(A2-a, 6 pts) Simplify Model 5 algebraically for occupied operation only (BH = 1) to the form $\hat{y} = m \cdot T + b$:

```
ANSWER_A2a_occ_slope = ____  
ANSWER_A2a_occ_intercept = ____  
print(f"ANSWER_A2a_occ_slope = {ANSWER_A2a_occ_slope}")  
print(f"ANSWER_A2a_occ_intercept = {ANSWER_A2a_occ_intercept}")
```

(A2-b, 6 pts) The ASHRAE 90.4 design condition is the **0.4% outdoor dry-bulb temperature = 103.8°F**. Using the occupied equation from A2-a, predict cooling demand at this design condition, and compare to a model using the Week 9 all-hours slope (1.1585 kWh/°F, intercept = -60.68 kWh):

```
ANSWER_A2b_pred_design = ____  
ANSWER_A2b_wk9_pred = ____  
print(f"ANSWER_A2b_pred_design = {ANSWER_A2b_pred_design}")  
print(f"ANSWER_A2b_wk9_pred = {ANSWER_A2b_wk9_pred}")  
  
A2b_diff = round(ANSWER_A2b_pred_design - ANSWER_A2b_wk9_pred, 2)  
print(f"A2b_diff = {A2b_diff}")
```

A2b_diff is deliberately not an ANSWER_ variable — it is there for your written response, and the two values it is built from are what get graded.

In a markdown cell, in one sentence: explain physically why the interaction model's occupied slope (≈ 1.78 kWh/°F) is steeper than the all-hours slope (≈ 1.16 kWh/°F).

(A2-c, 6 pts) The loop has **200 tons** (703.4 kWh/hr) of uncommitted plant capacity. Using the occupied prediction equation from A2-a, solve for the outdoor temperature at which this one facility would consume all of it:

```
plant_spare_kWh = 200 * 3.517  
ANSWER_A2c_Tsat = ____  
print(f"ANSWER_A2c_Tsat = {ANSWER_A2c_Tsat}")
```

In a markdown cell (2–3 sentences): is this temperature physically plausible for Tucson (max observed $\approx 106^\circ\text{F}$)? What does the answer tell the capital planning committee about whether this connection alone justifies funding new plant capacity — and what would have to change about the load for that conclusion to flip?

Track B — ArcE: ASHRAE 90.1 Occupied/Unoccupied Energy Model (18 pts)

ASHRAE Standard 90.1-2022 requires that energy compliance simulations model **occupied and unoccupied operating modes separately**. The interaction model gives you separate equations for each mode.

(B2-a, 6 pts) Simplify Model 5 into two separate equations — one for occupied hours (BH = 1), one for unoccupied hours (BH = 0):

```

ANSWER_B2a_occ_slope = ____
ANSWER_B2a_occ_intercept = ____
ANSWER_B2a_unocc_slope = ____
ANSWER_B2a_unocc_intercept = ____
print(f"ANSWER_B2a_occ_slope = {ANSWER_B2a_occ_slope}")
print(f"ANSWER_B2a_occ_intercept = {ANSWER_B2a_occ_intercept}")
print(f"ANSWER_B2a_unocc_slope = {ANSWER_B2a_unocc_slope}")
print(f"ANSWER_B2a_unocc_intercept = {ANSWER_B2a_unocc_intercept}")

```

(B2-b, 6 pts) ASHRAE 90.1 minimum-efficiency provisions include a **nighttime setback** requirement: the HVAC system must be capable of reducing cooling output by at least 30% during unoccupied periods. At the **1% design condition (T = 108°F)**:

```

ANSWER_B2b_occ_108 = ____
ANSWER_B2b_unocc_108 = ____
ANSWER_B2b_ratio = ____
print(f"ANSWER_B2b_occ_108 = {ANSWER_B2b_occ_108}")
print(f"ANSWER_B2b_unocc_108 = {ANSWER_B2b_unocc_108}")
print(f"ANSWER_B2b_ratio = {ANSWER_B2b_ratio}")

```

In a markdown cell: does the unoccupied load represent **less than 70%** of the occupied load at this temperature (i.e., is ANSWER_B2b_ratio < 0.70)? State your conclusion about whether this satisfies the setback intent.

(B2-c, 6 pts) Compute the **occupancy load fraction** at three outdoor temperatures:

$$\text{Occupancy fraction} = \frac{\hat{y}_{\text{occupied}} - \hat{y}_{\text{unoccupied}}}{\hat{y}_{\text{occupied}}}$$

```

def occ_fraction(T):
    occ = ANSWER_B2a_occ_slope*T + ANSWER_B2a_occ_intercept
    unocc = ANSWER_B2a_unocc_slope*T + ANSWER_B2a_unocc_intercept
    return (occ - unocc) / occ

```

```

ANSWER_B2c_frac_70 = ____
ANSWER_B2c_frac_90 = ____
ANSWER_B2c_frac_108 = ____
print(f"ANSWER_B2c_frac_70 = {ANSWER_B2c_frac_70}")
print(f"ANSWER_B2c_frac_90 = {ANSWER_B2c_frac_90}")
print(f"ANSWER_B2c_frac_108 = {ANSWER_B2c_frac_108}")

```

In a markdown cell: does the occupancy fraction **increase, decrease, or remain approximately constant** as outdoor temperature rises? Explain physically why: as outdoor temperature increases, does the incremental effect of occupancy become a larger or smaller fraction of total cooling demand?

Problem 3 — Written Reflection (12 pts, written)

Write a **100–150 word** reflection (no code required) in a markdown cell, addressing the following prompt:

The interaction model ($R^2 = 0.795$) outperforms the additive model ($R^2 = 0.697$) at the cost of greater complexity and a harder-to-explain coefficient structure. Describe **one real-world scenario** where you would recommend the simpler additive model despite its lower R^2 , and **one scenario** where the interaction model is the better choice. Support each recommendation with a **specific quantitative observation** from the exercise (e.g., a coefficient value, an R^2 comparison, a prediction difference, or a slope difference).

Grading criteria (12 pts total): - Additive model recommendation: scenario is specific and plausible, supported by at least one exercise number (4 pts) - Interaction model recommendation: scenario is specific and plausible, supported by at least one exercise number (4 pts) - Writing quality: concise, organized, quantitative evidence used appropriately, within 100–150 words (4 pts)

Submission Checklist

- ☐ Setup cell run with MAJOR declared
- ☐ P1a: two ANSWER_ season means printed + written response ~75 words using “collinearity”
- ☐ P1b: model summary shown; ANSWER_ r2/coef/pvalue printed; written comparison to Model 2
- ☐ P1c: model summary shown; ANSWER_ r2/adj_r2 printed; written conclusion about DayType’s incremental information
- ☐ P1d: four Model 5 coefficients stated; ANSWER_ predictions printed; written comparison to Model 2
- ☐ P2 (only your declared track): all ANSWER_ values printed and written responses
- ☐ P3: reflection, 100–150 words, two scenarios, two quantitative observations